

Q.30 Define: (CO4)

i) Skip Distance

ii) Multiple Hop propagation

Q.31 Explain Virtual Height in detail, (CO4)

Q.32 Explain the concept of Sky wave propagation. (CO4)

Q.33 Write the difference between active and passive satellites. (CO5)

Q.34 What are the conditions for Geostationary Satellite? (CO5)

Q.35 Draw the block diagram of Satellite communication system. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Draw & explain in detail communication Receiver. (CO2)

Q.37 Explain the characteristics and applications of Yagi-Uda Antenna. (CO3)

Q.38 Explain:

i) Pre-emphasis & De-emphasis (CO2)

ii) Characteristics and applications of VSAT. (CO5)

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4th Sem / Eltx, Power Eltx

Subject:- Communication Systems / Comm. Engg.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 FM transmitters are used for broadcasting in _____ range: (CO1)

a) V.H.F

c) Both a & b

b) U.H.F

d) None of these

Q.2 Super Heterodyne is a type of _____ (CO2)

a) Transmitter

c) Both a & b

b) Receiver

d) None of these

Q.3 The Intermediate frequency is equal to _____ (CO2)

a) $f_o - f_s$

c) $f_s - f_o$

b) $f_o + f_s$

d) None of these

Q.4 Frequency range of VHF is _____ (CO3)

a) 3-30 MHz

c) 300-3000 MHz

b) 30-300 MHz

d) 30-300 kHz

Q.5 Polarization may be classified as _____ (CO3)

a) Linear

c) Elliptical

b) Circular

d) All of above

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Q.6 Half-wave Dipole Antenna is also known as _____ (CO3)

- a) Power Antenna c) Loop Antenna
b) Hertz Antenna d) None of these

Q.7 LOS stands for _____ (CO4)

- a) Light of Sun c) Line of Sight
b) Line of Sun d) Light of Sight

Q.8 Layers of Ionosphere are _____ (CO4)

- a) D, E, F layers c) B, C, D layers
b) A, B, C layers d) All of above

Q.9 Critical frequency is given by _____ (CO4)

- a) $\frac{9}{\overline{N_{\max}}}$ c) $\frac{\overline{N_{\max}}}{9}$
b) $9\overline{N_{\max}}$ d) None of these

Q.10 An orbit is usually _____ (CO5)

- a) Square c) Both a & b
b) Elliptical d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Short-wave transmitter operates in which frequency range (CO1)

Q.12 Expand RADAR. (CO1)

Q.13 Define Sensitivity. (CO2)

Q.14 Expand AGC. (CO2)

Q.15 Define Beam width. (CO3)

Q.16 Define Antenna. (CO3)

Q.17 What is Critical frequency? (CO4)

Q.18 Expand MUF? (CO4)

Q.19 VSAT stands for _____ (CO5)

Q.20 What do you mean by Apogee? (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Draw the High Level AM Transmitter. (CO1)

Q.22 Write any five applications of Transmitters. (CO1)

Q.23 Define the following (CO2)

- i) Fidelity ii) S/N Ratio

Q.24 Write the selection criteria for intermediate frequency. (CO2)

Q.25 Draw the block diagram of FM Receiver. (CO2)

Q.26 Explain the concept of Polarization of EM waves. (CO3)

Q.27 Explain Folded Dipole Antenna. (CO3)

Q.28 Briefly explain the characteristics and applications of Half-Wave Dipole Antenna. (CO3)

Q.29 Explain Ground wave propagation. (CO4)